

POSTS AND TELECOMMUNICATIONS INSTITUTE OF TECHNOLOGY



PROJECT

Course: Foundation Internship

Lecturer: Nguyễn Trọng Khánh

Group: 04

Members:

Trịnh Thu Hồng B23DCVT180

Bùi Ngọc Thạch B23DCDT232

Lương Đức Thái B23DCVT387

Topic: A Study on Vision-Language Models
(VLMs) applied to Data Extraction

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1. Abstract	3
2. Introduction	4
3. Related Work	7
4. Methodology.....	9
4.1. Overall System Pipeline.....	9
4.2. Fine-Tuned Donut Model.....	10
4.4. Training Strategy	15
4.5. Inference Pipeline	20
4.6. OCR Baseline System.....	27
5. Experiments	29
5.1. Experimental Environment	29
5.2. Evaluation Metrics.....	30
5.3. Testing Procedure	30
5.4 OCR Baseline Evaluation	31
6. Results	31
6.1. Quantitative Results of the Fine-Tuned Donut Model	31
6.2. Qualitative Prediction Analysis	32
6.3 OCR Baseline Comparison	33
6.4. Failure Cases and Limitations	35
7. Discussion and Conclusion	37
8. Demo system	38
9. References	41

1. Abstract

Automatic invoice information extraction plays an important role in financial management, expense tracking, and Intelligent Document Processing (IDP) systems. Traditional invoice extraction methods typically rely on Optical Character Recognition (OCR) combined with rule-based processing techniques. Although OCR-based approaches are widely used, they often face difficulties when processing invoices with noisy backgrounds, inconsistent layouts, blur, or low image quality. Recent Vision-Language Models (VLMs) provide new approaches for document understanding by directly learning structured information from document images. Among these models, Donut (Document Understanding Transformer) is an OCR-free Vision-Language Model designed for end-to-end document information extraction.

This research proposes an OCR-free invoice information extraction system using the Donut model. The proposed system extracts important invoice fields including store name, transaction date, and total amount directly from invoice images and converts the extracted information into structured JSON outputs.

In this project, the pretrained `naver-clova-ix/donut-base` model is fine-tuned on a custom invoice dataset containing approximately 200 invoice images. The dataset is divided into training, validation, and testing subsets. During preprocessing, invoice annotations are converted into structured tag-based sequences compatible with Donut sequence generation.

The model is trained using Google Colab with NVIDIA T4 GPU acceleration. Experimental evaluation includes training loss analysis, validation loss monitoring, edit distance evaluation, and qualitative prediction analysis on unseen invoice samples. In addition, a traditional OCR-based baseline system using Tesseract OCR and regex-based extraction is implemented for comparison purposes.

Experimental results show that the fine-tuned Donut model achieves an Average Edit Distance of 5.30 and a Total Accuracy of 88.89% on the testing dataset. The qualitative evaluation further demonstrates that the proposed OCR-free approach can successfully extract structured invoice information from visually diverse invoice layouts.

To demonstrate practical applicability, the fine-tuned model is integrated into a Streamlit-based expense management web application for automatic invoice processing.

The results of this research indicate that OCR-free Vision-Language Models have strong potential for intelligent invoice understanding and financial document automation tasks.

Keywords: Invoice Information Extraction, Vision-Language Model, Donut, OCR-Free Extraction, Intelligent Document Processing, Expense Management.

2. Introduction

The rapid growth of digital transformation in finance and enterprise management has significantly increased the demand for automated document understanding systems. Among various business documents, invoices represent one of the most important document types used in accounting, expense management, taxation, and financial auditing workflows. Organizations process large volumes of invoices every day, making manual invoice handling time-consuming, expensive, and highly prone to human error. Consequently, automatic invoice information extraction has become an important research topic within Intelligent Document Processing (IDP) and Document AI.

Traditional invoice extraction systems are primarily based on Optical Character Recognition (OCR) technologies. In conventional pipelines, document images are first processed using OCR engines to extract textual content before downstream

algorithms identify key information fields such as invoice number, transaction date, vendor name, or total amount. Popular OCR systems such as Tesseract OCR, Google Vision API, and Amazon Textract have demonstrated strong text recognition capabilities under controlled conditions. However, OCR-based approaches still face multiple limitations when applied to real-world invoices.

Real invoice documents often contain highly diverse layouts, inconsistent formatting styles, multilingual text, noisy backgrounds, shadows, low-resolution scans, skewed images, and handwritten annotations. Errors occurring during OCR recognition frequently propagate to subsequent information extraction stages, reducing overall system reliability. Furthermore, many template-based extraction approaches require manually designed rules and layout-specific engineering, making them difficult to generalize across unseen invoice formats.

Recent advances in deep learning and Transformer architectures have significantly improved document understanding research. Vision-Language Models (VLMs) integrate visual and textual representations to perform complex document understanding tasks such as key information extraction, document classification, visual question answering, and structured sequence generation. Models such as LayoutLM, Pix2Struct, Qwen2-VL, Gemini, and Donut demonstrate strong capabilities in multimodal reasoning and document semantic understanding.

Among these approaches, Donut (Document Understanding Transformer) is particularly attractive for invoice processing because it eliminates dependence on external OCR engines. Donut formulates document understanding as an end-to-end sequence generation task using a Vision Encoder-Decoder architecture. Instead of performing explicit OCR recognition, the model directly converts document images into structured sequence representations, which are later parsed into JSON outputs for applications. This OCR-free paradigm simplifies the document processing pipeline while improving robustness against OCR-related recognition errors.

Several recent studies further demonstrate the growing importance of OCR-free and Vision-Language-based document understanding systems. Hybrid Vision-Language approaches integrating Donut with large language reasoning models such as Gemini have shown promising performance for intelligent invoice understanding and semantic reasoning. Other studies on bilingual medical invoice extraction also indicate that Vision-Language Models can effectively generalize across multilingual invoice layouts and complex financial document structures. Additionally, recent research on efficient document VLMs emphasizes the importance of OCR-free document parsing for improving scalability, robustness, and semantic consistency in practical document AI systems.

Motivated by these developments, this research proposes an OCR-free invoice information extraction framework using the Donut Vision-Language Model. The proposed system focuses on extracting three important invoice fields:

- Store name
- Transaction date
- Total amount

The extracted information is generated as structured sequence representations directly from invoice images and later converted into JSON fields. During training, invoice annotations are serialized into tag-based sequence representations compatible with Donut sequence generation. During inference, generated outputs are post-processed into structured JSON fields for the expense management system. The overall system is integrated into a Streamlit-based expense management application to demonstrate practical deployment for financial automation.

The main objectives of this research are summarized as follows:

1. Develop an OCR-free invoice extraction pipeline using the Donut model.
2. Fine-tune Donut on a custom invoice dataset with structured annotations.

3. Evaluate extraction performance using loss curves, edit distance metrics, and qualitative prediction analysis.
4. Analyze the strengths and limitations of OCR-free Vision-Language Models for invoice understanding.
5. Develop a practical demo system for automated invoice processing and expense management.

The contributions of this project can be summarized as follows:

- An end-to-end OCR-free invoice extraction framework is implemented using the Donut Vision-Language Model.
- A structured invoice dataset preparation pipeline is developed using tag-based sequence annotations for Vision-Language Model training.
- Experimental evaluation demonstrates the effectiveness of the proposed OCR-free invoice extraction approach on some invoice samples.
- A Streamlit-based web application is developed to demonstrate practical financial document automation.
- The research provides qualitative analysis of model behavior on visually diverse and noisy invoice layouts.

3. Related Work

3.1. OCR-Based Document Information Extraction

Traditional invoice information extraction systems mainly rely on Optical Character Recognition (OCR) technologies combined with rule-based information extraction methods. In conventional OCR pipelines, document images are first converted into machine-readable text using OCR engines before downstream modules identify important information fields.

After OCR processing, extracted text is analyzed using rule-based extraction techniques to obtain structured information such as store names, transaction dates, invoice numbers, and payment amounts.

The general OCR-based document extraction workflow is illustrated below.

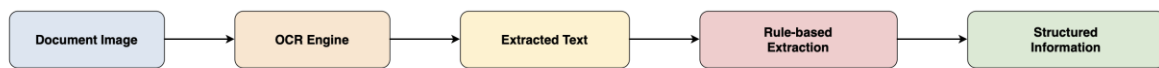


Figure show General OCR-based document information extraction pipeline.

OCR-based approaches are widely used in document automation tasks due to their simplicity and accessibility. These methods can achieve acceptable performance under controlled document layouts and high-quality image conditions.

However, OCR systems still face several limitations in real-world invoice processing scenarios. Invoice documents often contain noisy backgrounds, inconsistent layouts, blur, shadows, low-resolution scans, and visually complex formatting structures. OCR recognition errors may propagate to downstream extraction stages, reducing overall system reliability.

In addition, many OCR-based extraction systems require handcrafted rules and layout-specific customization, making them difficult to generalize across diverse invoice templates and unseen document structures.

Motivated by these limitations, recent studies have explored OCR-free Vision-Language Models for end-to-end document understanding and structured information extraction.

3.2. OCR-Free Vision-Language Models

Recent advances in Vision-Language Models (VLMs) have significantly improved document understanding research. Unlike traditional OCR-based approaches, OCR-free Vision-Language Models directly learn mappings between document images and structured outputs without requiring explicit text recognition stages.

Donut (Document Understanding Transformer) is one of the most representative OCR-free document understanding models. Donut adopts a Vision Encoder-Decoder architecture consisting of a Swin Transformer encoder and a Transformer decoder for sequence generation.

The OCR-free design simplifies document processing pipelines and reduces dependency on external OCR systems and handcrafted post-processing rules.

Recent studies have demonstrated the effectiveness of Donut and other Vision-Language Models in various document understanding tasks such as receipt parsing, invoice extraction, and structured document generation.

Motivated by these advances, this research adopts Donut as the primary architecture for invoice information extraction and compares it with a traditional OCR-based baseline system.

4. Methodology

4.1. Overall System Pipeline

This research proposes an OCR-free invoice information extraction system using the Donut Vision-Language Model. The proposed framework is designed to automatically extract important financial information from invoice images and convert the extracted information into structured JSON outputs for expense management applications.

The overall system consists of several stages, including invoice preprocessing, Vision-Language inference, structured sequence generation, JSON conversion, and web-based visualization.

The workflow begins with invoice image input uploaded through the Streamlit web application. The uploaded image is preprocessed using the Donut processor before being passed into the fine-tuned Donut model. The model then generates structured

textual sequences containing invoice information fields such as store name, transaction date, and total payment amount.

The generated sequences are subsequently post-processed and converted into structured JSON outputs for downstream expense management functionalities.

The complete system pipeline is illustrated below:



4.2. Fine-Tuned Donut Model

The proposed system uses Donut (Document Understanding Transformer), an OCR-free Vision-Language Model designed for end-to-end document understanding tasks.

The model adopts a Vision Encoder-Decoder architecture consisting of:

- a Swin Transformer encoder
- a Transformer decoder

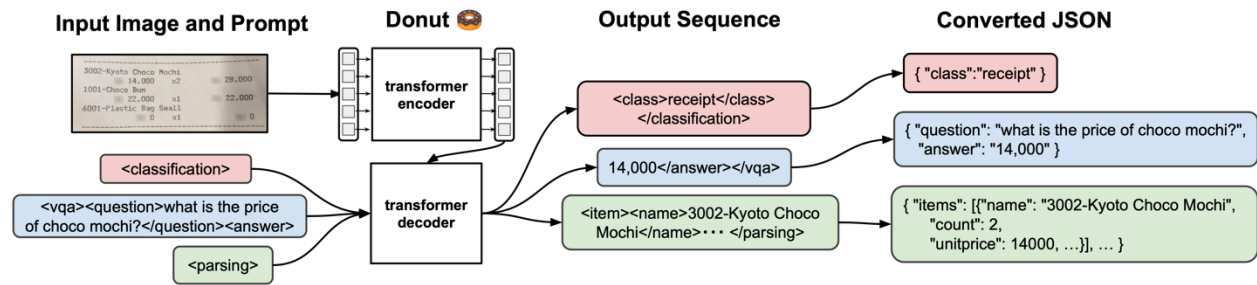


Figure X. Donut Vision Encoder-Decoder architecture adapted from Kim et al. (2022).

The encoder extracts visual representations directly from invoice images, while the decoder autoregressively generates structured textual outputs token by token.

Unlike conventional OCR pipelines, Donut eliminates explicit OCR stages and directly learns mappings between invoice images and structured outputs. This OCR-

free design simplifies the extraction pipeline and reduces dependency on OCR recognition quality.

In this research, the parsing capability of Donut is utilized for invoice information extraction. the pretrained naver-clova-ix/donut-base model is fine-tuned on a custom invoice dataset containing approximately 200 invoice images to generate structured invoice sequences.

The fine-tuned model is trained to extract three important invoice fields:

- store name
- transaction date
- total amount

During training, invoice annotations are serialized into structured tag-based sequences compatible with Donut sequence generation.

Example training sequence:

```
<s_invoice><store>VinCommerce</store><date>2020-08-14</date><total>52700</total></s_invoice>
```

During inference, generated outputs are post-processed and converted into structured JSON fields for downstream applications.

4.3. Dataset Preparation

The project uses a custom invoice dataset consisting of training, validation, and testing subsets. The dataset contains approximately 200 invoice images collected from multiple invoice layouts and formatting styles.

Each invoice image is associated with structured annotations containing:

- store name
- transaction date
- total amount

The annotations are stored in JSON format and later converted into structured training sequences for Donut fine-tuning.

Example annotation format:

```
{  
  "image": "invoice171.jpg",  
  "ground_truth": {  
    "store": "VinCommerce",  
    "date": "2020-08-14",  
    "total": "52700"  
  }  
}
```

The structured annotations are transformed into tag-based sequences during preprocessing to enable autoregressive sequence generation.

Example sequence representation:

```
<s_invoice><store>VinCommerce</store><date>2020-08-  
14</date><total>52700</total></s_invoice>
```

The dataset includes invoices with varying visual quality conditions, noisy backgrounds, and diverse formatting structures in order to improve model generalization capability.

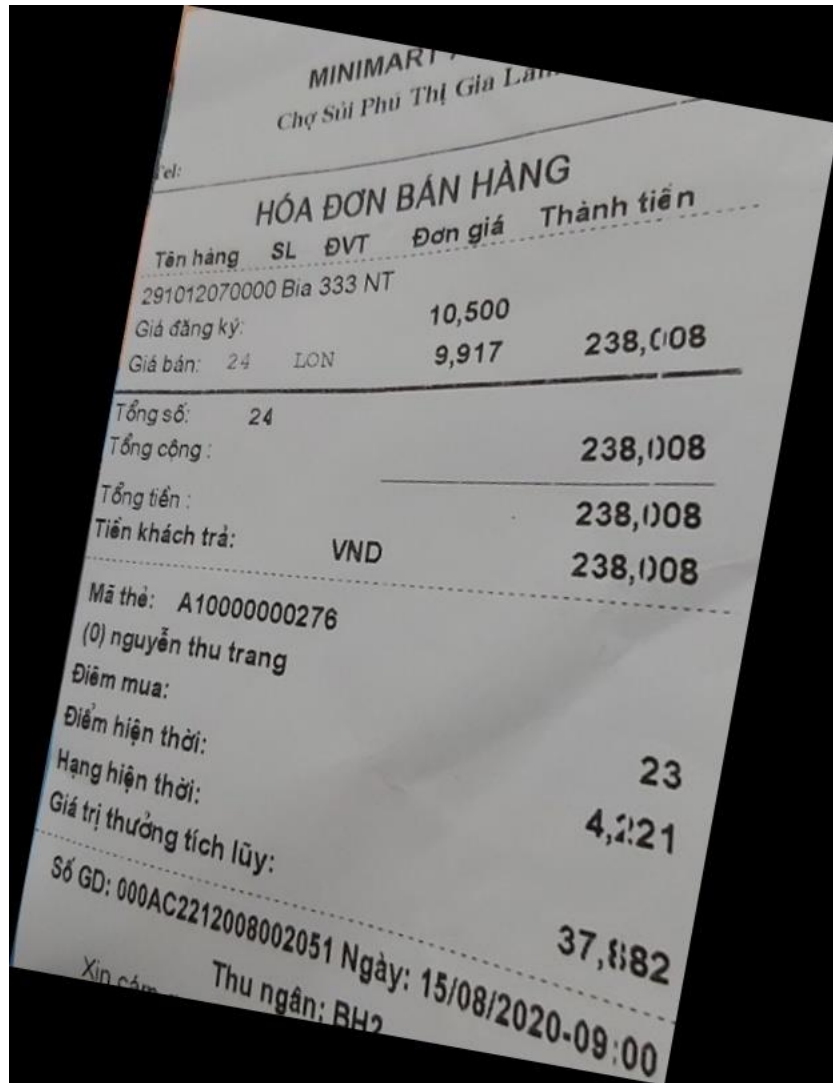
Some samples images:



Example invoice images in dataset/train

```
213     "image": "invoice26.jpg",
214     "ground_truth": {
215         "gt_parse": {
216             "store": "VM+ QNH Dự án KDC lân biển cọc 6",
217             "date": "2020-08-15",
218             "total": "122400"
219         }
220     }
221 },
```

Example structured annotation used in the invoice dataset.



Example invoice images in dataset/train

```

992 {
993   "image": "invoice105.jpg",
994   "ground_truth": {
995     "gt_parse": {
996       "store": "MINIMART ANAN",
997       "date": "2020-08-15",
998       "total": "238008"
999     }
1000   }
1001 },

```

Example structured annotation used in the invoice dataset.

The custom invoice dataset is divided into three subsets: training, validation, and testing. Approximately 70% of the dataset is allocated for training, 15% for validation, and 15% for testing.

The training set is used to fine-tune the Donut model and optimize model parameters during the learning process. The validation set is used to monitor model generalization performance during training and evaluate metrics such as validation loss and edit distance after each epoch. The testing set is reserved for final evaluation and qualitative prediction analysis on unseen invoice samples.

This dataset split strategy helps reduce overfitting and ensures that the trained model can generalize to visually similar invoice layouts that were not directly observed during training.

4.4. Training Strategy

The Donut model is fine-tuned using Google Colab with NVIDIA T4 GPU acceleration. The implementation is developed using PyTorch and HuggingFace Transformers.

The **pretrained naver-clova-ix/donut-base** model is used as the initialization checkpoint before fine-tuning on the custom invoice dataset.

The training configuration includes:

- batch size: 1
- training epochs: 5
- optimizer: AdamW
- learning rate: $3e-5$
- maximum sequence length: 512

Gradient checkpointing is enabled during training to reduce GPU memory consumption and improve training feasibility within the limited GPU memory environment available in Google Colab.

The model is trained using supervised sequence generation learning. During training, invoice images are passed through the Donut processor and converted into visual

embeddings. The decoder then autoregressively generates structured invoice sequences based on ground-truth annotations.

The training objective minimizes sequence generation loss between generated outputs and target structured sequences.

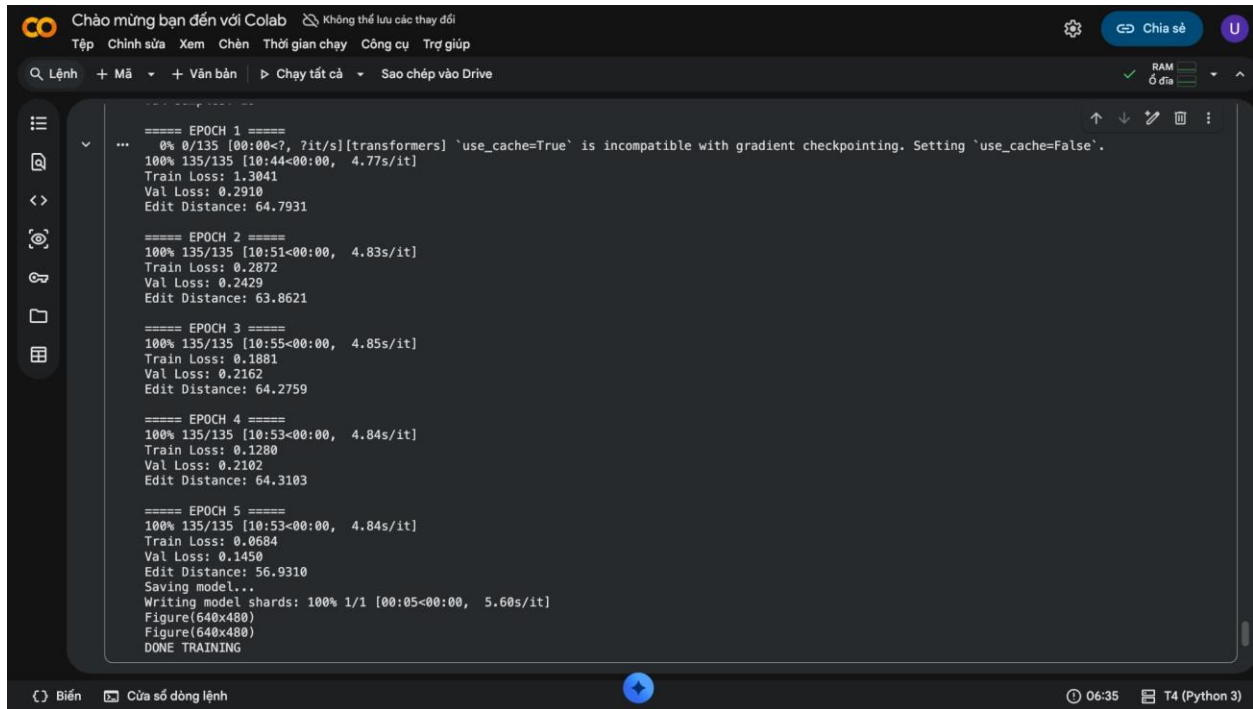
To evaluate training performance and model convergence, three evaluation metrics are monitored throughout the fine-tuning process:

- training loss
- validation loss
- edit distance

Training loss measures optimization performance on the training dataset, while validation loss evaluates the model's generalization capability on unseen validation samples.

Edit distance is additionally used to measure character-level similarity between generated outputs and ground-truth annotations. Lower edit distance values indicate better prediction quality and higher similarity between predicted and target sequences.

The experimental training results across five epochs are summarized in below table:



```
===== EPOCH 1 =====
0% 0/135 [00:00<?, ?it/s][transformers] `use_cache=True` is incompatible with gradient checkpointing. Setting `use_cache=False`.
100% 135/135 [10:44<00:00, 4.77s/it]
Train Loss: 1.3041
Val Loss: 0.2910
Edit Distance: 64.7931

===== EPOCH 2 =====
100% 135/135 [10:51<00:00, 4.83s/it]
Train Loss: 0.2872
Val Loss: 0.2429
Edit Distance: 63.8621

===== EPOCH 3 =====
100% 135/135 [10:55<00:00, 4.85s/it]
Train Loss: 0.1881
Val Loss: 0.2162
Edit Distance: 64.2759

===== EPOCH 4 =====
100% 135/135 [10:53<00:00, 4.84s/it]
Train Loss: 0.1280
Val Loss: 0.2102
Edit Distance: 64.3103

===== EPOCH 5 =====
100% 135/135 [10:53<00:00, 4.84s/it]
Train Loss: 0.0684
Val Loss: 0.1450
Edit Distance: 56.9310
Saving model...
Writing model shards: 100% 1/1 [00:05<00:00, 5.60s/it]
Figure(640x480)
Figure(640x480)
DONE TRAINING
```

Epoch	Train Loss	Validation Loss	Edit Distance
1	1.3041	0.2910	64.7931
2	0.2872	0.2429	63.8621
3	0.1881	0.2162	64.2759
4	0.1280	0.2102	64.3103
5	0.0684	0.1450	56.9310

Table show Training and validation results during Donut fine-tuning.

The training process demonstrates stable convergence throughout the fine-tuning procedure. Training loss decreases significantly from 1.3041 in the first epoch to 0.0684 in the final epoch, indicating that the model successfully learns invoice structure representations from the custom dataset.

Validation loss also gradually decreases across epochs, demonstrating improved generalization capability on unseen validation samples.

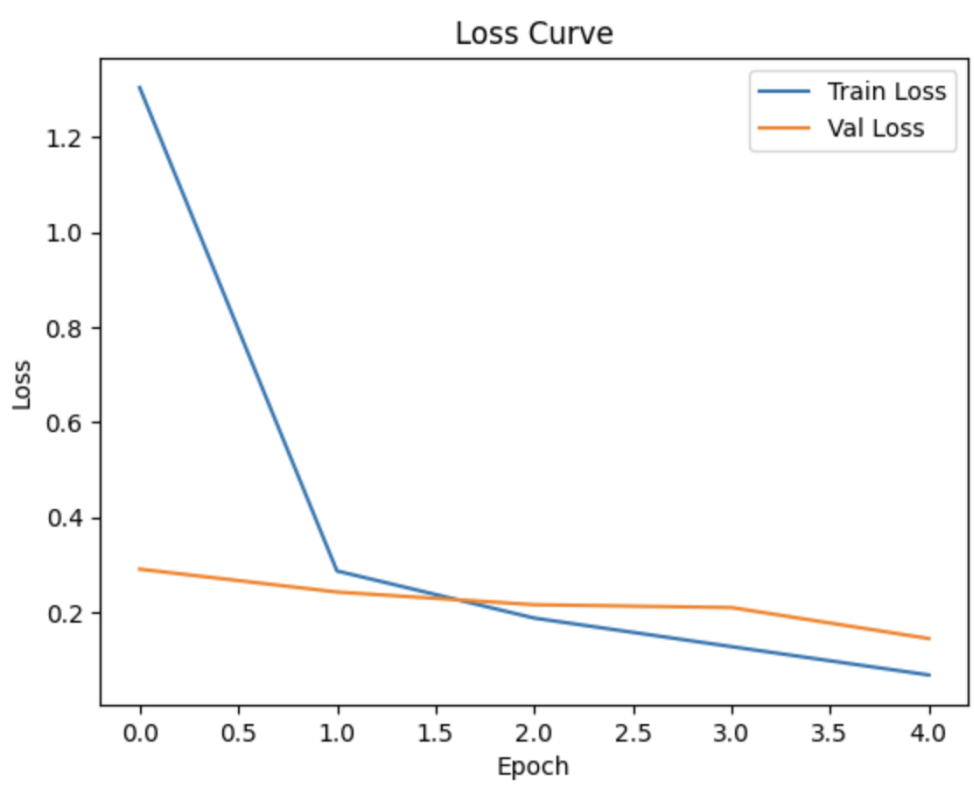


Figure show Training and validation loss curves during Donut fine-tuning.

Although edit distance fluctuates slightly during intermediate epochs, the metric improves significantly in the final epoch, decreasing from 64.7931 to 56.9310. This result indicates improved similarity between generated outputs and ground-truth invoice annotations.

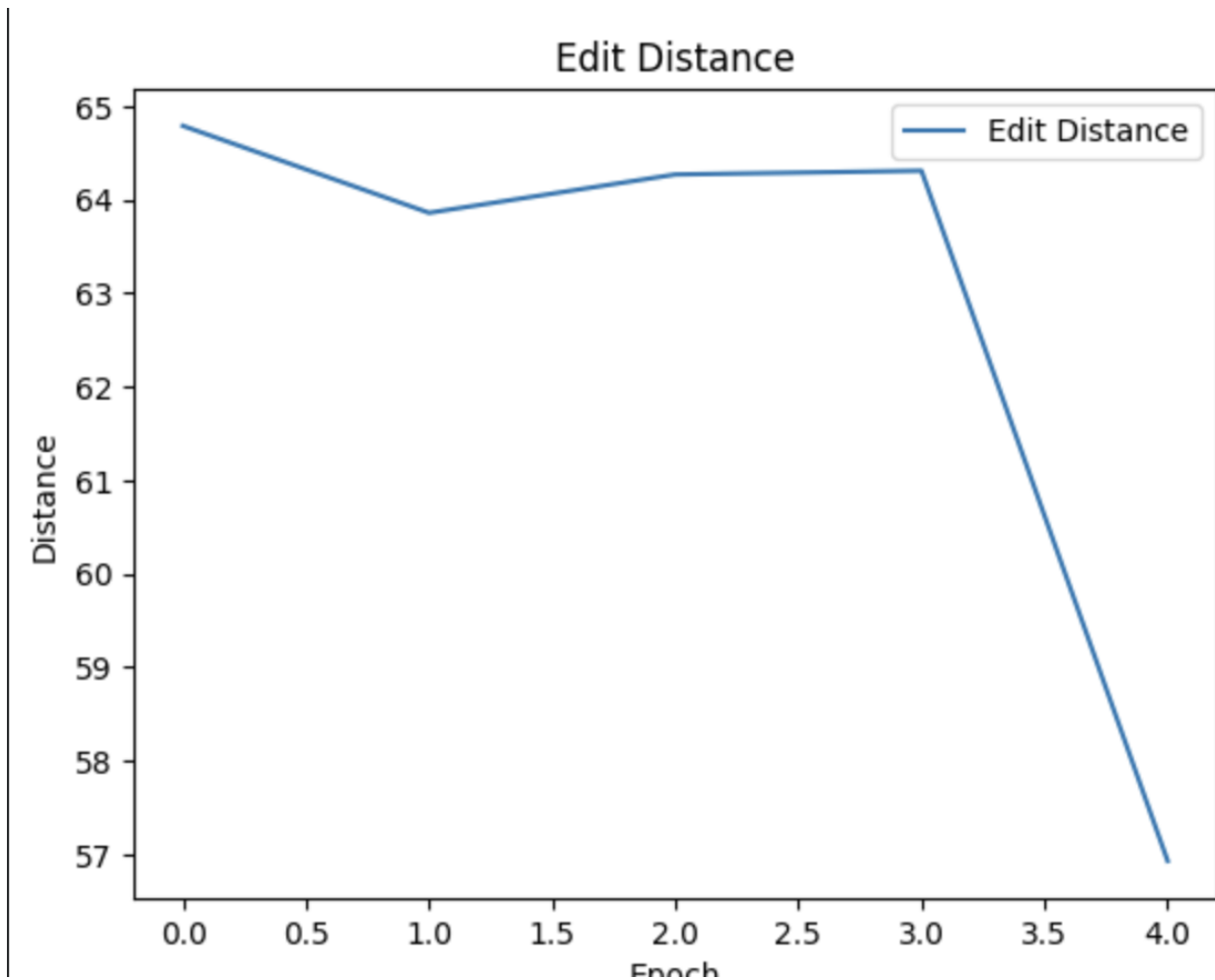


Figure show Validation edit distance across training epochs.

The training process requires approximately 10–11 minutes per epoch in the Google Colab T4 GPU environment.

After the completion of the training process, the fine-tuned Donut model is saved for downstream inference and deployment within the Streamlit expense management application.

The final model is subsequently evaluated on the unseen testing dataset containing 27 invoice images.

The final testing results are summarized below:

Metric	Result
Average Edit Distance	5.30
Total Accuracy	88.89%

Table show Final evaluation results on the testing dataset

The testing results demonstrate that the fine-tuned Donut model achieves strong invoice extraction performance on unseen invoice samples. The relatively low average edit distance and high total accuracy indicate that the model can successfully generalize to visually similar invoice layouts beyond the training dataset.

4.5. Inference Pipeline

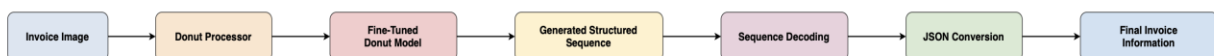
After the fine-tuning process is completed, the trained Donut model is used for invoice information extraction during inference. The inference stage is responsible for generating structured invoice information from unseen invoice images uploaded by users.

The inference pipeline is implemented in `donut_infer.py` using HuggingFace Transformers and PyTorch.

The inference process begins by loading the fine-tuned Donut model together with the corresponding Donut processor. The processor performs image preprocessing operations including image resizing, normalization, padding, and tensor conversion before passing the processed image into the Vision Encoder-Decoder model.

The fine-tuned model used during inference is the custom-trained version of **naver-clova-ix/donut-base** obtained after five epochs of fine-tuning on the invoice dataset.

The complete inference workflow is illustrated below:



During inference, invoice images are first converted into pixel embeddings using DonutProcessor. The processed image tensors are then passed into the Vision Encoder-Decoder architecture.

The decoder generates invoice information autoregressively using sequence generation methods. Instead of predicting all fields simultaneously, the decoder predicts tokens sequentially based on previously generated tokens and visual document embeddings.

The generated output follows the structured tag-based format used during training.

Example generated sequence:

```
<s_invoice><store>VinCommerce</store><date>2020-08-14</date><total>52700</total></s_invoice>
```

The generated sequence is subsequently decoded and post-processed into structured JSON fields for downstream applications.

Example JSON output:

```
{
  "store": "VinCommerce",
  "date": "2020-08-14",
  "total": "52700"
}
```

The JSON conversion stage extracts important invoice fields using structured tag parsing and converts them into machine-readable financial records.

Compared with traditional OCR pipelines, the proposed inference process directly generates structured invoice information in an end-to-end manner without requiring external OCR engines or handcrafted post-processing rules.

The inference pipeline is integrated into a Streamlit-based expense management web application. Users can upload invoice images through the web interface and visualize extracted invoice information in real time.

To evaluate model generalization capability, the fine-tuned Donut model is tested on unseen invoice images from the testing dataset.

The final testing results demonstrate:

- Average Edit Distance: 5.30
- Total Accuracy: 88.89%

These results indicate that the model can successfully generalize to visually similar invoice layouts beyond the training dataset.

The model demonstrates strong performance on numerical fields such as total amount and transaction date prediction. However, minor errors still occur in visually challenging invoices and noisy store name recognition cases.

Example prediction:

Prediction:

<s_invoice>...</s_invoice>

Ground Truth:

<s_invoice>...</s_invoice>

Some successful prediction examples:

Example 1:



invoice188.jpg

```
===== SAMPLE 6 =====  
IMAGE: invoice188.jpg  
PRED : <s_invoice><store>VinCommerce</store><date>2020-08-14</date><total>10000</total></s_invoice>  
GT : <s_invoice><store>VinCommerce</store><date>2020-08-14</date><total>10000</total></s_invoice>
```

Prediction Result

Example 2:



invoice191.jpg

```
===== SAMPLE 4 =====  
IMAGE: invoice191.jpg  
PRED : <s_invoice><store>VinCommerce</store><date>2020-08-14</date><total>52700</total></s_invoice>  
GT : <s_invoice><store>VinCommerce</store><date>2020-08-14</date><total>52700</total></s_invoice>
```

Prediction Result

Example 3:



Invoice199.jpg

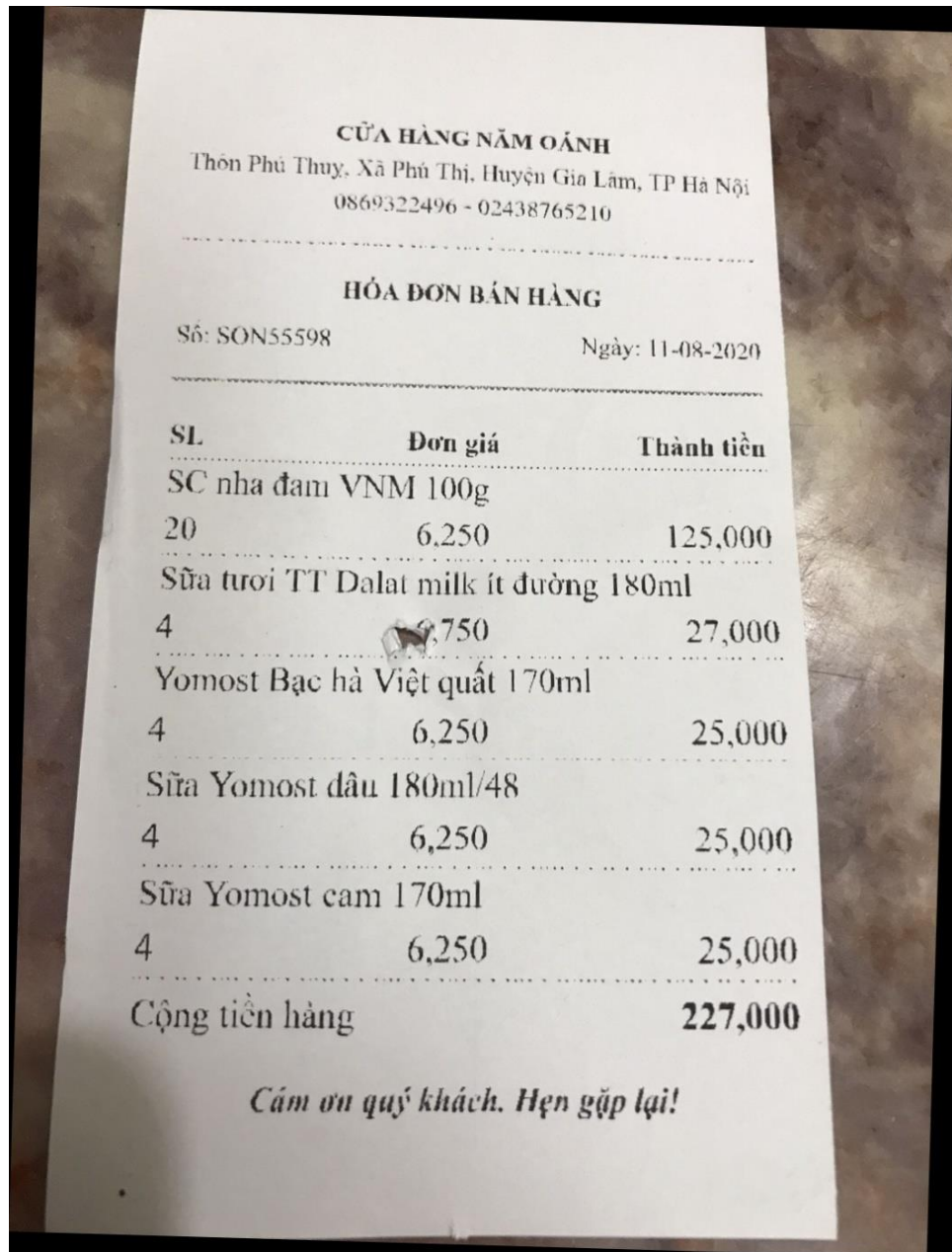
```
===== SAMPLE 7 =====
IMAGE: invoice199.jpg
PRED : <s_invoice><store>VinCommerce</store><date>2020-08-10</date><total>32272</total></s_invoice>
GT : <s_invoice><store>VinCommerce</store><date>2020-08-10</date><total>32272</total></s_invoice>
```

Prediction Result

The qualitative prediction examples demonstrate that the fine-tuned Donut model can successfully extract important invoice information fields from unseen invoice images. In several testing samples, the generated predictions exactly match the ground-truth annotations, particularly for structured numerical fields such as transaction dates and total payment amounts.

The results indicate that the model learns meaningful visual-textual representations from the custom invoice dataset and can generalize to visually similar invoice layouts during inference.

Partial Failure Example:



Invoice177.jpg

```
===== SAMPLE 8 =====  
IMAGE: invoice177.jpg  
PRED : <s_invoice><store>CỬA HÀNG NM OÁNH</store><date>2020-08-11</date><total>25000</total></s_invoice>  
GT   : <s_invoice><store>Cua Hang Nam Oanh</store><date>2020-08-11</date><total>227000</total></s_invoice>
```

Prediction Result

Despite the strong overall extraction performance, the fine-tuned Donut model still exhibits several limitations when processing visually challenging invoice samples. Figure above presents an example where the predicted total amount differs significantly from the ground-truth annotation.

In this sample, the model successfully recognizes the transaction date and partially identifies the store name. However, the predicted total amount is incorrectly generated as “25000” instead of the correct value “227000”. This error may occur due to noisy numerical regions, visually ambiguous character spacing, or complex invoice layouts that reduce numerical recognition accuracy.

The prediction result indicates that although the model learns meaningful visual-textual representations during fine-tuning, numerical extraction remains sensitive to image quality and layout complexity. In particular, invoices containing densely packed text, inconsistent formatting, or low visual clarity may still lead to incorrect sequence generation.

These limitations suggest that larger training datasets, more diverse invoice layouts, and longer fine-tuning strategies could further improve the robustness and generalization capability of the proposed OCR-free Vision-Language approach.

4.6. OCR Baseline System

To compare the effectiveness of the proposed OCR-free Vision-Language approach, a traditional OCR-based baseline system is implemented using Tesseract OCR and regex-based information extraction.

Unlike the proposed Donut model, the OCR baseline focuses only on extracting the total payment amount from invoice images.

The complete OCR baseline workflow is illustrated below:



Figure show OCR-based baseline pipeline for invoice total amount extraction.

During the OCR stage, invoice images are processed using Tesseract OCR to generate raw textual outputs from document images. The OCR implementation is developed in `ocr_infer.py`.

The extracted OCR text is subsequently analyzed using Vietnamese payment-related keywords and regular expressions implemented in `extract_total.py`.

Example payment-related keywords include:

- “tổng tiền”
- “tổng cộng tiền thanh toán”
- “total payment”

The system searches for these keywords inside OCR-generated text before extracting nearby numerical values using regex-based pattern matching.

Example OCR output:

Tong cong tien thanh toan: 52700

Example extracted result:

Total Amount = 52700

The OCR baseline is computationally lightweight and relatively simple to implement. In addition, it does not require GPU fine-tuning or large-scale model training.

However, the OCR pipeline still suffers from several limitations.

First, the extraction quality heavily depends on OCR recognition accuracy. If the OCR engine incorrectly recognizes invoice text due to blur, shadows, low image quality, or noisy layouts, downstream regex extraction may fail.

Second, the baseline system relies heavily on handcrafted keyword rules and predefined extraction logic. As a result, the system struggles to generalize across visually diverse invoice layouts and unseen formatting styles.

Third, the OCR baseline only extracts the total payment amount, whereas the proposed Donut model can simultaneously generate multiple invoice fields including store name, transaction date, and total amount in an end-to-end manner.

Compared with the OCR baseline, the proposed Donut-based Vision-Language approach provides:

- end-to-end invoice understanding
- OCR-free structured generation
- multi-field extraction capability
- improved robustness against noisy invoice layouts

The comparison between the OCR baseline and the fine-tuned Donut model demonstrates the advantages of Vision-Language Models for intelligent document understanding and financial document automation.

5. Experiments

5.1. Experimental Environment

The Donut Vision-Language Model is fine-tuned using Google Colab with NVIDIA T4 GPU acceleration. The implementation is developed using Python, PyTorch, and HuggingFace Transformers.

The OCR baseline system is implemented separately using Tesseract OCR together with regex-based extraction methods.

The Streamlit framework is additionally used to develop the invoice expense management demonstration interface.

5.2. Evaluation Metrics

To evaluate the effectiveness of the proposed OCR-free Vision-Language approach, several quantitative evaluation metrics are used during training and testing.

The evaluation metrics include:

- Training Loss
- Validation Loss
- Edit Distance
- Total Extraction Accuracy

Training loss and validation loss are used to monitor model convergence during fine-tuning.

Edit distance measures character-level similarity between generated outputs and ground-truth annotations. Lower edit distance values indicate higher prediction quality.

Total extraction accuracy evaluates whether the predicted invoice total exactly matches the ground-truth annotation.

In addition to quantitative evaluation, qualitative prediction analysis is also conducted using unseen invoice samples from the testing dataset.

5.3. Testing Procedure

After fine-tuning, the trained Donut model is evaluated on unseen invoice images from the testing dataset.

During testing, invoice images are processed through the inference pipeline implemented in `donut_infer.py`. The generated outputs are compared with ground-truth annotations to calculate edit distance and extraction accuracy.

Several unseen invoice samples are selected for qualitative prediction analysis during the evaluation process.

The OCR baseline system is also evaluated on the same testing samples for comparison purposes.

5.4 OCR Baseline Evaluation

The OCR baseline system is evaluated on several invoice samples to qualitatively compare OCR-based extraction with the proposed Donut approach.

The OCR baseline uses:

- Tesseract OCR
- keyword matching
- regex-based numerical extraction

Unlike the Donut model, the OCR baseline focuses only on extracting the total payment amount.

Experimental observations show that the OCR baseline performs correctly on simple and visually clean invoice samples. However, the extraction pipeline becomes less reliable under noisy layouts, low image quality, or inconsistent invoice formatting.

In several testing examples, the OCR baseline successfully extracts the correct payment amount in approximately half of the evaluated samples, while the remaining cases contain OCR recognition or regex extraction errors.

These observations further motivate the use of OCR-free Vision-Language Models for robust invoice understanding.

6. Results

6.1. Quantitative Results of the Fine-Tuned Donut Model

The final testing results demonstrate that the fine-tuned Donut model achieves strong invoice extraction performance on unseen invoice samples.

The model achieves:

- Average Edit Distance: 5.30
- Total Accuracy: 88.89%

The relatively low edit distance indicates strong similarity between generated outputs and ground-truth annotations.

The high total extraction accuracy further demonstrates the effectiveness of the proposed OCR-free Vision-Language approach for invoice understanding tasks.

6.2. Qualitative Prediction Analysis

Several qualitative prediction examples demonstrate the practical inference capability of the fine-tuned Donut model on unseen invoice images.

In multiple testing samples, the generated predictions exactly match the ground-truth annotations, particularly for numerical fields such as transaction dates and total payment amounts.

The model successfully extracts structured invoice information from visually diverse invoice layouts without relying on external OCR engines.



invoice184.jpg

PRED: <s_invoice><store>VinCommerce</store><date>2020-08-13</date><total>9500</total></s_invoice>

GT: <s_invoice><store>VinCommerce</store><date>2020-08-13</date><total>9500</total></s_invoice>

Edit Distance: 0

6.3 OCR Baseline Comparison

The OCR baseline system demonstrates acceptable performance on several simple invoice samples with visually clean layouts. However, the extraction quality strongly depends on OCR text recognition accuracy and predefined keyword rules.

In the conducted qualitative evaluation, the OCR baseline successfully extracts correct payment amounts for approximately half of the tested invoice samples, while other cases contain OCR recognition or regex matching errors.

Compared with the OCR baseline, the fine-tuned Donut model demonstrates:

- end-to-end structured generation
- multi-field extraction capability
- improved robustness against layout variations
- reduced dependency on handcrafted extraction rules

These results indicate the advantages of OCR-free Vision-Language Models for intelligent invoice understanding tasks.

- OCR success example

```
===== Đang xử lý: invoice175.jpg =====  
Tổng tiền (Total) tìm được: 145000  
-----
```

```
===== Đang xử lý: invoice189.jpg =====  
Tổng tiền (Total) tìm được: 345000  
-----
```

- OCR failure example

```
===== Đang xử lý: invoice188.jpg =====  
Tổng tiền (Total) tìm được: 9021213  
-----
```

- Invoice: invoice188.jpg
- Correct Total Amount: 10,000 VND
- OCR Output: 9,021,213 VND

```
===== Đang xử lý: invoice176.jpg =====  
Tổng tiền (Total) tìm được: 3021213  
-----
```

- Invoice: invoice176.jpg

- Correct Total Amount: 8,435 VND
- OCR Output: 3,021,213 VND
- Correct output total extracted by Donut

```
[6/27] invoice176.jpg
RAW   : <s_invoice><store>VinCommerce</store><date>2020-08-14</date><total>8435</total></s_invoice>
PARSED: {'store': 'VinCommerce', 'date': '2020-08-14', 'total': '8435'}
```

```
[17/27] invoice188.jpg
RAW   : <s_invoice><store>VinCommerce</store><date>2020-08-14</date><total>10000</total></s_invoice>
PARSED: {'store': 'VinCommerce', 'date': '2020-08-14', 'total': '10000'}
```

6.4. Failure Cases and Limitations

Despite the strong overall extraction performance, the proposed model still exhibits several limitations when processing visually challenging invoice samples.

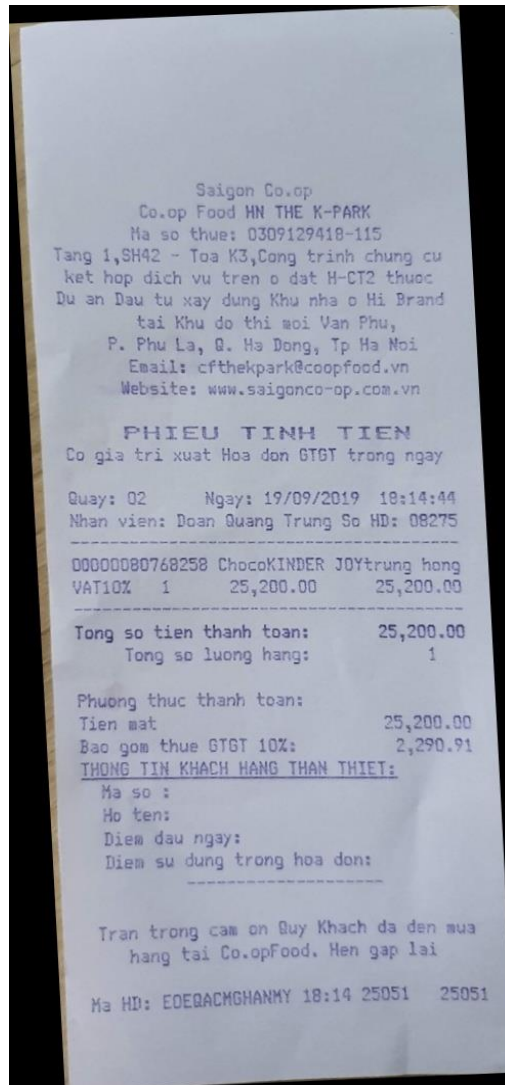
One representative failure case occurs when the model incorrectly predicts the invoice total amount due to noisy numerical regions and visually ambiguous layouts. In this example, the model successfully recognizes the transaction date and partially identifies the store name. However, the predicted total amount differs significantly from the ground-truth annotation.

These prediction errors indicate that numerical extraction remains sensitive to image quality, invoice formatting complexity, and limited dataset diversity.

Future improvements may include:

- larger invoice datasets
- more diverse invoice layouts
- longer fine-tuning strategies
- improved numerical token handling

Example:



Invoice193.jpg

Result:

PRED: <s_invoice><store>Saigon Co.op</store><date>2019-09-19</date><total>25200000</total></s_invoice>

GT : <s_invoice><store>Saigon Co.op</store><date>2019-09-19</date><total>25200</total></s_invoice>

Edit Distance: 3

7. Discussion and Conclusion

This research presents an OCR-free invoice information extraction system using the Donut Vision-Language Model.

The proposed approach fine-tunes the pretrained naver-clova-ix/donut-base model on a custom invoice dataset containing approximately 200 invoice images. The system is designed to extract structured invoice information including store name, transaction date, and total payment amount directly from invoice images.

Experimental results demonstrate that the fine-tuned Donut model achieves strong extraction performance on unseen invoice samples, obtaining:

- Average Edit Distance: 5.30
- Total Accuracy: 88.89%

The qualitative evaluation further shows that the proposed OCR-free approach provides improved robustness against visually diverse invoice layouts compared with the traditional OCR-based baseline system.

Unlike the OCR baseline, which depends heavily on OCR text quality and handcrafted extraction rules, the proposed Donut-based approach performs end-to-end structured information generation without requiring external OCR processing. The experimental observations additionally indicate that the fine-tuned Donut model successfully learns meaningful visual-textual representations from the custom invoice dataset and can generalize to visually similar unseen invoice layouts.

However, several limitations are still observed during experimentation.

First, the dataset size remains relatively small, containing approximately 200 invoice images. Limited dataset diversity may reduce generalization capability for highly complex invoice structures or visually different layouts.

Second, prediction errors occasionally occur in visually noisy invoice samples, particularly in dense numerical regions and low-quality invoice images.

Third, the OCR baseline comparison in this research remains relatively limited and qualitative. More extensive benchmarking using additional OCR systems and larger evaluation datasets could provide more comprehensive comparative analysis.

Despite these limitations, the proposed system demonstrates the practical applicability of OCR-free Vision-Language Models for intelligent invoice understanding and automated expense management applications.

Future work may include:

- expanding the invoice dataset
- increasing layout diversity
- exploring larger Vision-Language Models
- improving numerical extraction robustness
- integrating semantic post-processing techniques

Overall, this research demonstrates the potential of OCR-free Vision-Language Models for intelligent financial document automation and structured invoice information extraction.

8. Demo system

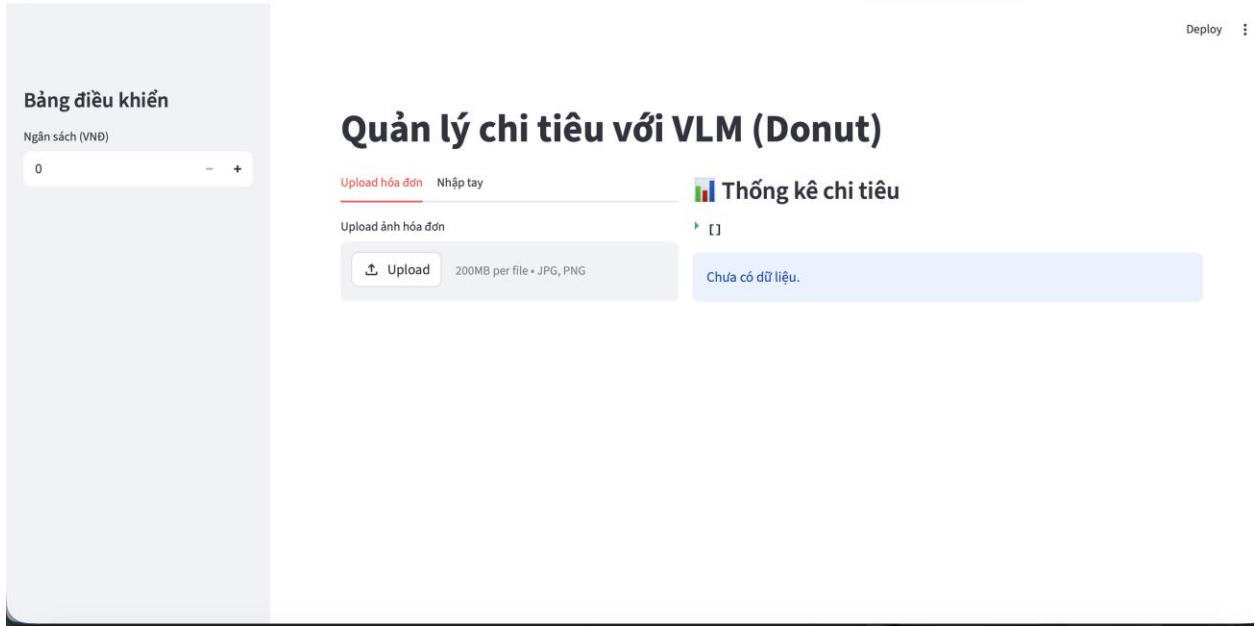
Link github: <https://github.com/hong0110/vision-language-model>

The demo system workflow can be divided into three major stages:

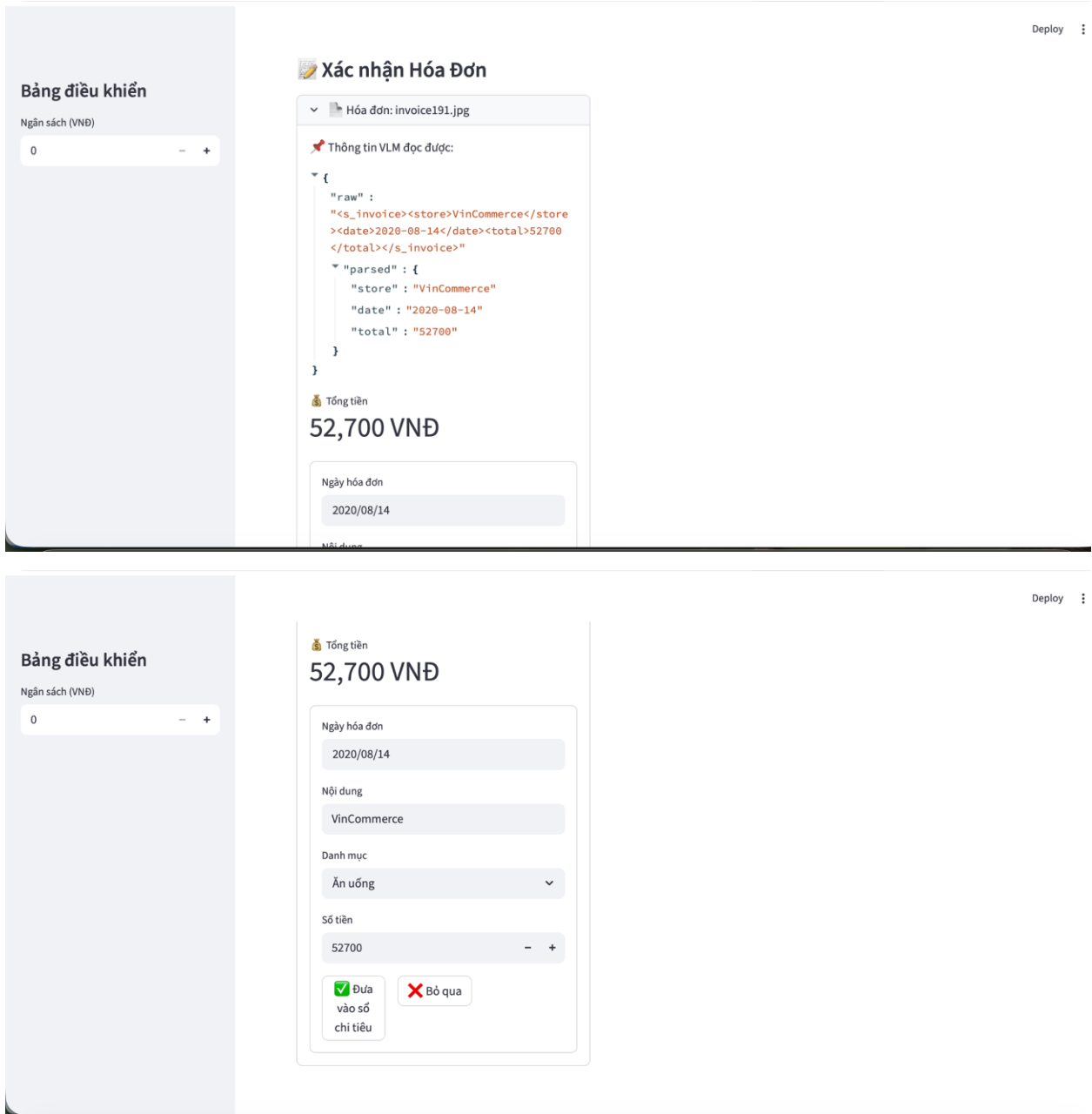
- **Initial application interface:** The Streamlit-based expense management system provides invoice upload functionality together with expense tracking and

visualization

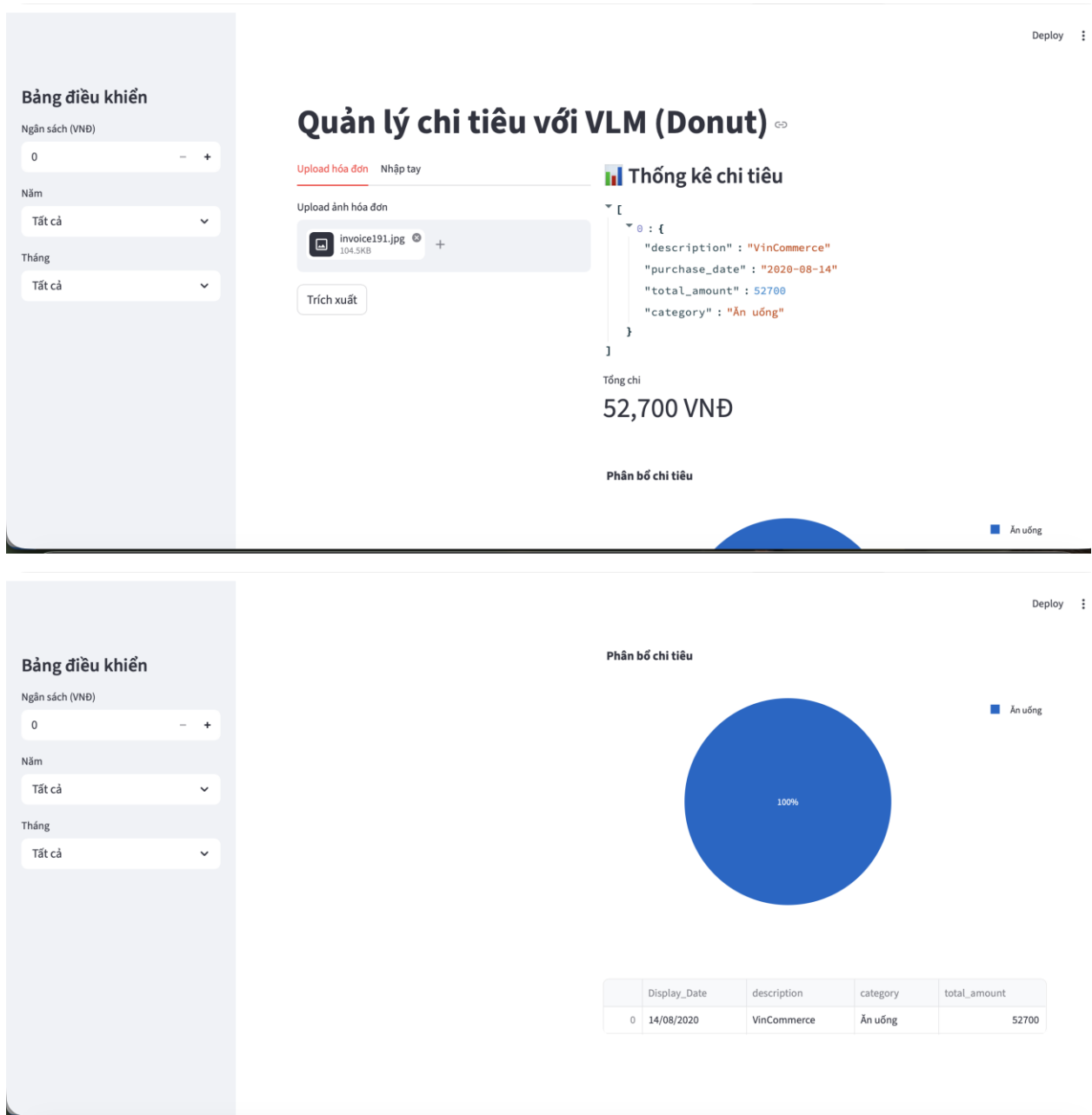
components.



- **Automatic invoice extraction and confirmation:**After an invoice image is uploaded, the fine-tuned Donut model automatically extracts structured invoice information including store name, transaction date, and total payment amount. The extracted information is then displayed for user verification before storage.



- **Expense tracking and visualization:**After confirmation, extracted invoice records are added into the expense management dashboard, where users can analyze spending statistics, categorized expenses, and transaction history through charts and structured tables.



9. References

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